

BE-SAFE: INITIAL DEVELOPMENTS OF A DATA-DRIVEN FRAMEWORK FOR CYCLIST SAFETY IN SMART CITIES

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ABSTRACT

Cyclist safety is a growing concern in smart cities, where the promotion of active mobility was not always followed by equally adaptation of infrastructure, traffic management, and digital systems (Santos et al., 2022). The BE-SAFE project (Bicycle SAFETy for Mobility and Active Riding in Smart Cities) addresses this gap by developing an integrated, data-driven framework to assess cycling risk and support evidence-based safety interventions. Methodologically, BE-SAFE integrates GIS-based accident analysis, machine-learning techniques for identifying key accident-contributing factors, cyclist behavioural profiling, IoT/V2X real-time monitoring, and simulation-based safety assessment within a unified decision-support framework. Building on the project's initial developments, this work presents the first results of four core pillars: accident data analysis and modelling, cyclist psychology and behaviour, design of a cyclist monitoring system using sensors on bikes for future laboratory and urban testing, and the simulation of cyclist-vehicle interactions under different behavioural conditions. In Task 1, historical bicycle-vehicle crash data from the Lisbon Metropolitan Area (LMA, Portugal) are compiled and analysed using GIS tools and emerging hotspot analysis to identify spatial-temporal clusters of accident occurrence and to develop preliminary machine learning models for identifying key risk factors. In Task 2, an online Cyclist Behaviour Survey covering violations, errors, positive behaviours, risk perception, perceived infrastructure safety, near misses, and socio-demographics, has been developed, translated from English into four languages (Portuguese, Romanian, Italian, Spanish), approved by the universities ethics committees, and pilot-tested across partner universities. In Task 3, the project

designed an IoT-based cyclist monitoring system for deployment on bike-sharing systems, designed to collect data on location, speed, acceleration, weather, braking patterns, and traffic proximity, while testing Vehicle-to-Everything (V2X) communication integration within a smart city platform. Finally, in Task 4, the integrated dataset was translated into behavioural cyclist profiles and incorporated into a SUMO-based simulation framework. This framework was employed to analyse the safety implications of different cyclist behaviour patterns under mixed-traffic conditions, using surrogate safety measures and traffic performance indicators to support the assessment of future urban mobility and infrastructure scenarios.

Results from the previous tasks are integrated into a cyclist safety framework that connects accident spatial intelligence, behavioural analysis, sensing technologies, and simulation outputs within a unified Cyclist Safety Knowledge Hub, supporting data-driven decision-making and laying the foundation for future applications aimed at safer and more sustainable urban mobility. The expected contribution of this work is: (i) a multi-source evidence base on cyclist risk patterns in different European contexts, (ii) a modular, transferable decision-support framework for municipalities and road safety authorities, and (iii) initial policy-oriented recommendations and guidelines to promote safer and more attractive cycling in smart cities.

1. INTRODUCTION

1.1 Context and Research Gap

Cycling is increasingly recognised as a key component of sustainable urban mobility strategies, contributing to the reduction of greenhouse gas emissions, improved public health, and the creation of more liveable cities (Logan et al., 2023, European Commission, 2025). In recent years, many European cities have invested in cycling infrastructure, traffic calming measures, and policies aimed at promoting active travel. Despite these efforts, cyclist safety remains a major challenge and continues to represent a significant barrier to the wider adoption of cycling as a daily mode of transport (Winters et al., 2012).

Road safety studies have consistently shown that cyclists are among the most vulnerable road users, particularly in urban environments characterised by high traffic volumes, complex intersections and street conditions, and frequent interactions with motorised vehicles (Duren et al., 2023).

The growing availability of mobility-related data has created new opportunities to support evidence-based safety interventions. Accident databases, behavioural surveys, IoT devices, connected vehicle technologies, and traffic simulation platforms provide complementary sources of information that can improve the understanding of cyclist safety from multiple perspectives. However, current research on cyclist safety remains largely fragmented. Spatial analyses primarily focus on identifying crash hotspots and spatial risk patterns (Jaber & Csonka, 2023; Panwinkler et al., 2024), behavioural studies seek to explain cyclists' decision-making processes and risk perception (Winters et al., 2012, Cafiso et al., 2021), sensor-based approaches concentrate on real-time monitoring of cyclist exposure and interactions (Li et al., 2025; Martinez et al., 2024), while simulation models are commonly used to evaluate the safety implications of alternative traffic, infrastructure, and mobility scenarios (Losada et al., 2024; Ayad et al., 2024).

Although these individual approaches have generated valuable insights, their limited integration constrains the development of comprehensive decision-support systems capable of capturing the complexity of cyclist safety in urban contexts. While

substantial progress has been achieved in accident analysis, behavioural studies, sensing technologies and simulation-based safety assessment, only a limited number of studies have attempted to integrate these dimensions within a single analytical framework. Existing efforts such as RobotCycle (Panagiotaki et al., 2024), cyclist-oriented V2X safety assessment (Schories et al., 2024), infrastructure risk perception studies (Cafiso et al., 2021), and Digital Twin-based traffic safety environments (Wang et al., 2023) demonstrate the benefits of combining multiple data sources, yet fully integrated frameworks linking accident intelligence, cyclist behaviour, real-time monitoring and simulation remain scarce. Such integration is essential to move beyond reactive safety management towards predictive and proactive approaches that can support planning, operation, and policy decisions in smart cities.

To address this gap, the BE-SAFE (Bicycle SAFETY for Mobility and Active Riding in Smart Cities) project proposes an integrated framework that combines multiple sources of safety-related information to support the assessment, prediction, and mitigation of cyclist risk. The framework brings together accident analysis, behavioural intelligence, IoT-enabled monitoring, and simulation-based evaluation within a common architecture designed to support future Digital Twin applications for cyclist safety.

1.2 Objectives

This paper presents the initial developments of the BE-SAFE framework and discusses its potential as a data-driven decision-support tool for improving cyclist safety in smart city environments.

More specifically, the study pursues four objectives:

1. To develop an integrated conceptual framework that combines accident analysis, behavioural data, real-time monitoring, and traffic simulation into a unified approach for cyclist safety assessment.
2. To present the main methodological components of the framework, including accident pattern analysis, behavioural profiling, IoT-based monitoring architectures, and simulation-based safety evaluation.
3. To report the initial results obtained in the first stages of the project, highlighting how different data sources contribute to the identification and understanding of cyclist risk factors.
4. To discuss how the integration of these components can support future Digital Twin environments and evidence-based decision-making, enabling more predictive, proactive and adaptive cyclist safety strategies in urban areas.

By integrating descriptive, predictive, and simulation-based analyses within a common framework, the BE-SAFE project seeks to contribute to the development of next-generation safety management tools capable of supporting safer, more resilient, and more sustainable urban mobility systems.

2. THE BE-SAFE FRAMEWORK

Improving cyclist safety requires a comprehensive understanding of the multiple factors influencing crash occurrence, risk exposure, and interactions within complex traffic environments. While substantial research has advanced accident analysis, cyclist behaviour, sensor-based monitoring, and traffic simulation, limited integration across these domains continues to restrict the development of holistic cyclist safety assessment and decision-support approaches.

The BE-SAFE project addresses this challenge through a data-driven framework designed to connect different sources of knowledge within a common analytical ecosystem. The underlying premise is that cyclist safety cannot be fully understood through a single perspective. Instead, it requires the combination of historical accident data, behavioural information, real-time observations, and simulation-based experimentation. To achieve this objective, the project was organised around four complementary research tasks, each addressing a specific question related to cyclist safety.

The first task focuses on understanding accident patterns through the spatial analysis of cyclist accident data. By combining Geographic Information Systems (GIS), spatial analysis techniques, and machine learning models, this component seeks to identify where and when accidents occur, determine the factors associated with crash occurrence and severity, and produce dynamic distribution and trend maps that support the identification of priority intervention areas. This task provides the foundational knowledge required to understand the spatial and temporal distribution of cyclist safety problems.

The second task addresses the behavioural dimension of cyclist safety. Although accident records reveal the outcomes of unsafe situations, they provide limited information regarding the behavioural factors that contribute to risk exposure. Through large-scale cyclist surveys and behavioural profiling methods, this task aims to identify cyclist groups with different levels of risk exposure and to characterise the behavioural factors associated with these differences. The results provide insights into risk perception, near-miss experiences, behavioural violations, and positive safety practices, allowing the identification of cyclist profiles that can subsequently be incorporated into simulation and decision-making processes.

While the first two tasks primarily rely on historical records and self-reported data, the third task introduces a real-time observational perspective on cyclist safety using sensors, Internet of Things (IoT) technologies, and V2X communications. This component aims to capture dynamic conditions within the transport environment as they occur, enabling the continuous monitoring of cyclist behaviour, environmental conditions, and interactions with surrounding road users and infrastructure. Data collected from instrumented bicycles provide valuable insights into cyclist movement patterns, route characteristics, exposure to risk, and proximity-based interactions with vehicles and roadside elements. Furthermore, recent advances in connected mobility have demonstrated the potential of V2X technologies to enhance the safety of vulnerable road users, including cyclists, by enabling the real-time exchange of safety-critical information between bicycles, vehicles, infrastructure, and traffic management systems. These capabilities support cooperative awareness, improve situational understanding, and facilitate the early detection and mitigation of potential conflicts (Schories et al., 2024; Karoui et al., 2022).

The fourth task focuses on testing solutions through simulation and Digital Twin technologies. Using traffic simulation platforms, behavioural models and infrastructure scenarios can be evaluated under controlled conditions. This component addresses the question of what happens if behaviours, infrastructure characteristics, or technological solutions are modified. By reproducing different cyclist profiles, traffic conditions, and intervention strategies, simulation provides a virtual experimentation environment that supports the assessment of safety impacts before real-world implementation.

Although each task generates valuable knowledge individually, the true value of BE-SAFE emerges when these different perspectives are connected within a single

knowledge cycle. Patterns identified through accident analysis provide context for behavioural investigations; behavioural profiles inform simulation scenarios; real-time monitoring data support simulation model calibration and validation; and simulation results contribute to the evaluation of potential interventions. Together, these elements create an integrated process capable of transforming isolated analyses into a comprehensive cyclist safety intelligence framework – the Cyclist Safety Knowledge Hub.

Figure 1 presents the proposed integrated BE-SAFE framework. The framework places the Cyclist Safety Knowledge Hub at its centre, acting as a common layer where information generated by the four project components is combined. By integrating accident, behavioural, sensing, and simulation data, the knowledge hub provides a comprehensive understanding of cyclist safety and supports informed decisions on risk management, infrastructure planning, and policy design.

A key characteristic of the framework is the incorporation of a continuous learning and improvement feedback loop. New data collected through monitoring activities can be used to refine behavioural models, update risk maps, improve simulation scenarios, and enhance future decision-making processes. As a result, the framework functions as a dynamic learning environment capable of continuously adapting to evolving urban mobility patterns and conditions.



Figure 1 – BE-SAFE Integrated Cyclist Safety Framework. *Source: Developed by the authors using Microsoft Copilot (Generative AI), 2026.*

3. METHODOLOGY AND DATA SOURCES

The BE-SAFE framework integrates multiple data sources and analytical approaches to provide a comprehensive understanding of cyclist safety in urban environments.

3.1 Data Sources

The development of the BE-SAFE framework relied on four primary data sources corresponding to the main analytical components of the project.

The first dataset consists of cyclist crash records collected for the Lisbon Metropolitan Area (LMA). The accident database includes 3807 cyclist-related crashes recorded between 2017 and 2024 across the 18 municipalities of the metropolitan region. The dataset contains information regarding accident location, temporal characteristics, road environment, and injury severity, providing the basis for spatial and predictive safety analyses.

The second data source corresponds to an international cyclist survey conducted in Portugal, Spain, Italy, and Romania. A total of 740 valid responses were collected, covering socio-demographic characteristics, cycling frequency, risk perception, near-miss experiences, cycling behaviour, and attitudes towards safety-related technologies. The survey was designed to support the identification of behavioural patterns associated with different levels of cyclist risk exposure.

The third source of information derives from the design and evaluation of a sensing and communication framework intended to support the future collection of cyclist safety-related data. The proposed architecture integrates bicycle-mounted sensors, environmental sensing devices, IoT technologies, and V2X communications, creating the technological foundation for future real-time monitoring of cyclist–environment interactions in both laboratory and urban settings. Finally, a simulation environment was developed to reproduce cyclist–vehicle interactions under different behavioural and operational conditions.

The fourth source of information is generated through a simulation framework that combines SUMO (Simulation of Urban Mobility)¹ and VEINS (Vehicles in Network Simulation)² to reproduce cyclist–vehicle interactions under different behavioural and traffic conditions, enabling the generation of safety-related indicators and the evaluation of alternative urban mobility scenarios.

3.2 Analytical Methods

Different analytical methods were employed to extract knowledge from each data source and generate the information required by the integrated BE-SAFE framework.

Accident Pattern Analysis

The accident intelligence component combines Geographic Information Systems (GIS), spatial statistics, and machine learning techniques to identify cyclist accident patterns and contributing factors.

Spatial clustering and distribution patterns are assessed through Nearest Neighbour Analysis (NNA), providing an overall measure of accident concentration. Global Moran's I is subsequently applied to evaluate spatial autocorrelation and determine whether accident occurrences exhibited statistically significant clustering patterns. To identify areas of elevated cyclist accidents, Kernel Density Estimation (KDE) is employed to generate continuous pattern surfaces across the study area. Additionally, Emerging Hot Spot Analysis (EHSA) is employed to analyse the spatio-temporal evolution of cyclist crashes and detect statistically significant hotspot trends, revealing

locations where crash concentrations persisted, intensified, diminished, or emerged over a study period.

To complement the GIS-based analyses, forest-based and boosted machine-learning algorithms are used to model cyclist crash severity and identify the key factors associated with injury severity. In addition to supporting predictive analyses, these approaches were used to evaluate variable importance and support the interpretation of hotspot patterns and accident severity relationships.

The proposed methodology was validated using a database of bicycle–vehicle crashes recorded in the Lisbon Metropolitan Area (2017–2024), providing an initial assessment of the applicability of the integrated GIS- and machine learning-based framework for accident pattern analysis (Sandoval & Santos, 2026).

Behavioural Analysis and Cyclist Profiling

The behavioural component seeks to understand how cyclist characteristics and behaviours influence safety outcomes. Behavioural indicators are derived from responses associated with violations, riding errors, positive cycling behaviours, and near-miss experiences, following the principles of the Cyclist Behaviour Questionnaire (CBQ) and the Risk Perception and Regulation Scale (RPRS) (Useche et al., 2018; Useche et al., 2024).

Survey responses are analysed using descriptive statistical techniques to characterise cyclist demographics, travel habits, risk perception, and safety-related attitudes. To identify groups of cyclists with distinct behavioural characteristics, a K-means cluster analysis is performed using variables derived from the survey. The clustering procedure aims to segment cyclists according to their behavioural patterns and levels of risk exposure, thereby supporting the development of representative cyclist profiles (Van der Zwaard et al., 2019).

The resulting behavioural profiles are translated into cyclist-agent parameters for modelling different behavioural conditions and levels of risk exposure. This process enables the integration of behavioural insights into the assessment of cyclist vulnerability under alternative traffic and infrastructure scenarios within the simulation framework.

To validate the proposed approach, the CBQ and RPRS questionnaires, along with other scales were administered to urban cyclists from four European countries (Portugal, Romania, Italy, and Spain). Responses were recorded using five-point Likert scales and subsequently processed for behavioural profiling and comparative analysis.

Real-Time Monitoring Architecture

The observational component of BE-SAFE focused on the evaluation of sensing and communication solutions for future cyclist safety monitoring applications.

Different sensor configurations were analysed to support the collection of data on cyclist and bicycle dynamics, environmental conditions along cycling routes, and interactions with surrounding vehicles and urban infrastructure. This process informed the specification of a candidate sensing platform integrating bicycle-mounted motion sensors, environmental sensors, and edge-computing devices for local data acquisition and processing.

In parallel, alternative data communication technologies were assessed through coverage and capacity analyses considering different propagation conditions, transceiver configurations, and operational requirements. The evaluation included short-range, low-power, cellular, and next-generation communication technologies,

with the objective of defining a communication architecture capable of supporting reliable data transmission in urban cycling environments and future V2X-enabled applications.

Simulation and Safety Assessment

The simulation component established a virtual environment for analysing cyclist safety and testing alternative mobility and infrastructure scenarios under controlled conditions (Segata et al., 2017, Lo Cigno et al., 2022, Xu et al., 2025). An integrated framework based on SUMO–Veins is defined to model cyclist–vehicle interactions, incorporating behavioural profiles derived from the CBQ and calibrated using empirical evidence from the accident, survey, and monitoring analyses. These profiles are translated into cyclist agent parameters representing operational and behavioural characteristics such as speed, acceleration, reaction time, following distance, behavioural variability, and junction interactions.

Safety assessment is based on surrogate safety measures (SSMs), including Time-to-Collision (TTC), Post-Encroachment Time (PET), and Deceleration Rate to Avoid Collision (DRAC), which are used to evaluate potential conflicts and risk exposure without relying on observed crash occurrences (Mahmud et al., 2017, Thunberg et al., 2021). Particular attention is given to near-conflict indicators, enabling the simulation framework to capture cyclist–vehicle interactions and assess safety performance across different traffic and infrastructure conditions.

To validate the proposed approach, the framework was tested in a pilot case study involving a compact four-arm urban intersection, allowing the assessment of cyclist–vehicle interactions under different behavioural scenarios and operational conditions.

3.3 Framework Integration

The methodological outputs generated by the four components were subsequently integrated within the BE-SAFE framework presented in Section 2.

The accident spatial intelligence component contributed evidence on cyclist crash locations patterns, and the factors associated with crash occurrence and severity.

The behavioural component provided cyclist profiles derived from survey data, enabling the identification of distinct behavioural patterns and levels of risk exposure.

The observational component contributed for the specification of a sensing and communication architecture for monitoring cyclist and bicycle dynamics, environmental conditions, and interactions with surrounding vehicles and urban infrastructure, while also evaluating communication solutions capable of supporting future V2X-enabled applications.

The simulation component translated the behavioural profiles into calibrated cyclist-agent parameters and incorporated them into an integrated SUMO–Veins environment, enabling the representation of heterogeneous cyclist behaviours and the evaluation of cyclist–vehicle interactions under alternative traffic and infrastructure scenarios. Safety performance was assessed through surrogate safety measures, providing a proactive approach to the analysis of potential conflicts and risk exposure. These complementary sources of knowledge were consolidated within the Cyclist Safety Knowledge Hub, where accident spatial intelligence, behavioural analysis, sensing and communication specifications, and simulation models are combined within a unified decision-support framework. The resulting platform can support applications such as cyclist risk assessment, infrastructure planning and prioritisation, safety-oriented route analysis, and evidence-based policy development.

Furthermore, the integration of monitoring, behavioural, and simulation capabilities establishes the foundations for future adaptive and data-driven cyclist safety management systems, including digital twin applications for safer and more sustainable urban mobility

4. INITIAL RESULTS

4.1 Understanding Cyclist Risk

The spatial analysis of 3807 cyclist crashes recorded in the Lisbon Metropolitan Area revealed statistically significant clustering patterns, confirming that cyclist accidents are not randomly distributed across the study area. Results obtained through Kernel Density Estimation (KDE) and Emerging Hot Spot Analysis (EHSA) identified several persistent and emerging hotspots concentrated along major urban corridors, densely populated areas, and locations characterised by complex interactions between cyclists and motor vehicles. These findings highlight the existence of priority intervention areas where cyclist exposure and accident occurrence are consistently higher.

The temporal dimension of the analysis further revealed changes in hotspot intensity throughout the study period, indicating that bicycle accident occurrence is influenced not only by infrastructure characteristics but also by evolving mobility patterns and urban development processes. While some hotspot locations remained stable over time, others exhibited intensifying trends, suggesting the emergence of new safety challenges requiring targeted interventions.

Concerning the behavioural mechanisms, the survey was completed by 740 cyclists from four European countries: Portugal (n = 399), Romania (n = 171), Italy (n = 92), and Spain (n = 78).

The analysis consistently identified two cyclist profiles across all participating countries: a High-Risk profile, characterised by higher levels of violations and errors and lower engagement in protective behaviours, and a Safety-Oriented profile, characterised by lower levels of risky behaviours and higher levels of protective behaviours. The proportion of cyclists in each group was remarkably stable across countries, with approximately 31–41% classified as High-Risk and 59–69% as Safety-Oriented.

Based on the average distribution observed in the two largest samples (Portugal and Romania), the simulation framework adopted a representative split of approximately 30% High-Risk cyclists and 70% Safety-Oriented cyclists. These behavioural profiles were subsequently used to parameterise simulation scenarios and support the analysis of cyclist vulnerability under different traffic and infrastructure conditions.

The integration of accident spatial intelligence and behavioural profiling constitutes one of the main contributions of the BE-SAFE framework. While spatial analyses identify where accident patterns are concentrated, behavioural analyses help explain why some cyclists may be more vulnerable than others. Together, these results support the development of more targeted safety interventions that address both infrastructural and behavioural dimensions of cyclist safety. This integrated perspective forms the basis for the subsequent stages of monitoring and simulation within the framework.

4.2 Towards Real-Time Safety Monitoring

Building upon the understanding of cyclist risk obtained from accident and behavioural analyses, the next stage of the BE-SAFE framework seeks to establish the foundations

for real-time safety monitoring through the deployment of sensing, communication, and connectivity technologies.

A preliminary IoT and V2X architecture was developed to support the continuous collection of information from the cycling environment. The proposed architecture combines bicycle-mounted sensing devices, environmental monitoring systems, communication networks, and data integration mechanisms capable of supporting future real-time safety applications. The architecture was designed following interoperability and scalability principles, enabling future integration with broader smart city infrastructures.

The selected sensing system incorporates accelerometers, distance sensors, and other motion-related devices capable of capturing braking events, speed variations, cyclist manoeuvres, and proximity-based interactions with nearby vehicles and urban infrastructure. Environmental sensors measuring variables such as temperature, humidity, and atmospheric conditions were also considered, recognising the potential influence of environmental factors on cyclist behaviour and safety. Data collection and preliminary processing are supported through ESP32-based edge devices, providing a flexible and cost-effective technological solution for local sensing, data aggregation, and communication.

Several communication technologies were evaluated, including Wi-Fi, LoRaWAN, NB-IoT, LTE-M, and 5G networks. Each technology presents different advantages regarding coverage, energy consumption, latency, and deployment cost, allowing the framework to be adapted to different operational contexts. V2X communication capabilities were also incorporated into the framework design to support future interactions between cyclists, connected vehicles, and smart infrastructure.

Preliminary results indicated that LoRa/LoRaWAN solutions can support communication ranges exceeding 1 km, while Wi-Fi connectivity is generally limited to distances between approximately 10 to 200 m. Cellular technologies, including NB-IoT, LTE-M, LTE, and 5G NR FR1, were identified as suitable options for applications requiring extended coverage and higher mobility, with final performance depending on the deployment locations of gateways, access points, and base stations.

Although the proposed sensing and communication architecture has not yet been validated through field deployments, its design establishes the foundations for future real-time cyclist safety monitoring.

By enabling the acquisition and integration of real-time data, the proposed architecture provides a technological basis for dynamic risk assessment and the continuous refinement of cyclist safety models. These capabilities constitute an important step toward future implementations of the BE-SAFE framework, particularly in the development of Digital Twin applications, cooperative V2X services, and adaptive cyclist safety management strategies.

4.3 Scenario-Based Safety Assessment

The simulation framework was used to assess the safety implications of different cyclist behavioural profiles through a pilot application at a road intersection. Four scenarios were considered: a car-only baseline scenario, a Safety-Oriented cyclist scenario, a High-Risk cyclist scenario, and a mixed scenario reflecting the behavioural distribution identified in the survey (70% Safety-Oriented and 30% High-Risk cyclists). The results revealed clear differences in safety performance across the simulated scenarios. Crossing conflicts extracted through surrogate safety measures proved particularly effective for analysing cyclist–vehicle interactions and distinguishing between behavioural profiles. In contrast, longitudinal conflicts were found to be

strongly influenced by traffic exposure, queuing effects, and shared-lane following behaviour, making them less suitable for assessing intersection-related cyclist risk. The Safety-Oriented scenario consistently generated fewer safety-critical interactions, reflecting more cautious riding behaviour and lower exposure to potential conflicts. Conversely, the High-Risk scenario produced the highest number of critical interactions, with more frequent occurrences of low TTC and PET values, indicating increased collision risk. The mixed scenario produced intermediate results, reflecting the coexistence of different behavioural patterns within the traffic environment. The analysis also showed that crossing conflict indicators provide a more robust representation of cyclist safety performance at intersections than traditional collision outputs. While collision records were useful for identifying simulation anomalies, teleports, and modelling artefacts, surrogate safety measures offered a more reliable basis for evaluating near-miss events and comparing behavioural scenarios. Overall, the findings confirm that cyclist behaviour significantly influences interaction patterns and safety outcomes. More importantly, they demonstrate the capability of the simulation framework to assess the potential impacts of behavioural changes, infrastructure interventions, and traffic management measures before their implementation in real-world environments. These results provide an important step towards integrating behavioural intelligence into safety-oriented traffic simulations and support the broader objectives of the BE-SAFE framework.

5. DISCUSSION

The initial results demonstrate the complementary nature of the four analytical components developed within the BE-SAFE framework. Individually, each component contributes to a specific dimension of cyclist safety, collectively, they establish a continuous knowledge cycle capable of supporting evidence-based decision-making and future Digital Twin applications.

Table 1 summarises the main outputs obtained during the project.

The results indicate that the framework already contains the key building blocks required for the future development of a cyclist safety Digital Twin. The combination of historical data, behavioural models, real-time monitoring, and simulation-based experimentation creates the foundations for a continuously evolving system capable of supporting adaptive safety interventions in smart city environments.

The principal contribution of BE-SAFE is therefore not the individual analytical components but their integration within a common Cyclist Safety Knowledge Hub. This integration represents a progression from descriptive analyses towards predictive and proactive cyclist safety management.

Table 1. Summary of BE-SAFE outputs.

Component	Main findings	Contribution
Task 1 – Accident Spatial Intelligence	Spatial analyses revealed significant clustering of cyclist crashes and identified persistent and emerging hotspots. Machine learning models enabled the identification of factors associated with accident occurrence and severity.	Identifies where and when cyclist accident concentration occurs and supports the prioritisation of intervention areas and safety investments.
Task 2 – Behavioural Analysis	Two cyclist profiles were identified: Safety-Oriented Cyclists and High-Risk Cyclists. Significant differences were observed regarding risk perception, behavioural violations, near misses, and exposure to hazardous situations.	Explains who is more exposed to risk and provides behavioural parameters for simulation and safety assessment.
Task 3 – Real-Time Monitoring	An IoT- and V2X-enabled architecture was exploited, integrating bicycle-mounted sensors, environmental sensing devices, edge computing, and wireless communication technologies.	Provides real-time observations of cyclist behaviour and interactions, enabling continuous monitoring and future near-miss detection.
Task 4 – Simulation and Digital Twin	Safety-Oriented scenarios generated fewer critical conflicts than Mixed and High-Risk scenarios. TTC, PET, and DRAC indicators showed increasing conflict severity as cyclist risk-taking behaviour increased.	Enables the evaluation of behavioural, infrastructural, and technological interventions before real-world implementation.
Integrated BE-SAFE Framework	Accident spatial intelligence, behavioural profiling, real-time monitoring, and simulation outputs were integrated within a common Cyclist Safety Knowledge Hub supported by continuous feedback mechanisms.	Establishes the foundations for evidence-based decision-making and future cyclist safety Digital Twin applications.

6. CONCLUSIONS AND FUTURE DEVELOPMENTS

This paper presented the initial developments of the BE-SAFE framework, a data-driven approach designed to improve cyclist safety. Unlike traditional approaches that address safety dimensions separately, BE-SAFE combines multiple sources of information within a unified analytical framework, enabling a more comprehensive understanding of cyclist risk and supporting evidence-based decision-making.

The results obtained during the first phase of the project demonstrate the value of this integrated perspective. Spatial and machine-learning analyses identified significant cyclist crash clustering patterns and priority intervention areas, while behavioural profiling revealed distinct cyclist groups with different levels of risk exposure. In parallel, the development of a IoT and V2X monitoring architecture established the foundations for continuous observation of cyclist–vehicle interactions, and simulation experiments confirmed the influence of behavioural characteristics on safety outcomes through surrogate safety indicators.

A key contribution of the project is the creation of the Cyclist Safety Knowledge Hub, which connects the outputs of the different analytical components and transforms isolated datasets into actionable safety intelligence. Through this integration, the framework supports a progression from descriptive safety analysis towards predictive and proactive safety management, creating opportunities for improved risk assessment, infrastructure prioritisation, route planning, and policy development.

The work presented in this paper represents the first step towards a broader research vision that will continue through the BE-SAFE-Next (Bicycle SAFETY powered by AI and NEXT-gen Communication Toward Cognitive Cities) initiative. Building directly on the results of BE-SAFE, the new project expands the consortium with additional academic and socio-economic partners and extends the original framework by incorporating advanced artificial intelligence techniques, next-generation communication technologies, distributed data fusion, cognitive V2X systems, and real-world pilot demonstrations in the cities of Pamplona, Covilhã, and Brasov. The objective is to move from the current laboratory-validated framework towards operational validation in real urban environments, while transforming isolated safety analyses into a cognitive decision-support ecosystem for smart cities.

Future developments will focus on integrating dynamic pattern and risk maps, cyclist behavioural profiles, real-time sensor observations, and cognitive simulation environments within a common platform capable of supporting near-real-time safety assessment and intervention. Emphasis will be placed on the fusion of AI-based risk modelling, V2X communications, sensorized bicycle fleets, and simulation technologies to anticipate cyclist-vehicle conflicts before they occur and support adaptive safety management strategies. This evolution aligns with the emerging concept of cognitive cities, where connected infrastructure, intelligent systems, and human-centred interventions continuously learn from urban mobility data and support safer, more sustainable transport systems.

NOTES

¹ - <https://sumo.dlr.de/docs/index.html>

² - <https://veins.car2x.org/>

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